

Qunsheng Hou

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Research Interests: Memristive devices and neuromorphic hardware; analog in-memory computing; device-aware algorithms for robust and efficient continual learning.

Education

Southern University of Science and Technology (SUSTech), Shenzhen, China Expected Jul. 2027

B.Eng. in Microelectronics / GPA: 3.72/4.00

Selected coursework: Materials Science (94/100); Analog Integrated Circuits (93/100); GaN Semiconductor Materials and Devices (91/100); Digital Integrated Circuits (90/100).

Publication

- Li, Y.[†], Yang, J.[†], **Hou, Q.**[†], et al. “Adaptive Redox Resistive Memory Programming for Efficient and Robust Class-Incremental Learning.” *Advanced Materials*, 2026. [†]Co-first authors. DOI: 10.1002/adma.202600025.

Research Experience

Undergraduate Researcher, Zhongrui Wang’s Lab, SUSTech Jul. 2024–Present

Memristor-aware continual learning and hardware-aware computing

- Co-first author of an *Advanced Materials* study on adaptive programming of redox resistive memory for class-incremental learning; investigated programming stochasticity and selective updating of high-impact weights to reduce programming overhead and mitigate catastrophic forgetting.
- Designed and simulated a device-variability-aware programming scheme for two-dimensional memristor arrays in Python; developed robustness-testing workflows and PCA/confusion-matrix analyses for the publication.
- Investigated data-segmentation and robustness strategies for MobileNet-based class-incremental learning on CIFAR-100, achieving 93% peak accuracy in the evaluated setting.

Semiconductor Device Fabrication and Characterization, SUSTech Mar. 2026–May 2026

- Fabricated Ta/Al₂O₃-based semiconductor capacitors in a cleanroom using RCA cleaning, photolithography, LPCVD, ALD, Ta sputtering, ICP etching, and rapid thermal annealing.
- Performed C–V characterization and SEM-based failure analysis to examine the effect of tantalum oxidation on capacitance and device integrity.

Selected Technical Projects

VLA Manipulation and Residual Reinforcement Learning Mar. 2026–May 2026

- Led a four-member team investigating residual reinforcement learning for vision–language–action manipulation; conducted parameter-efficient fine-tuning of the 3-billion-parameter GR00T model.
- Developed and validated VLA manipulation pipelines on SO101 and Unitree G1-D platforms, including simulation validation, real-time inference, and support for real-world deployment.

RISC-V CPU Design Jul. 2025

- Led a three-member team in designing a five-stage pipelined RV32I CPU in Verilog with a 2-bit saturating-counter branch predictor and 4-way set-associative cache; verified it on a Xilinx Nexys A7 FPGA at 185 MHz.

Technical Skills and Honors

Programming/ML: Python; deep learning; continual learning; CNNs; Mamba and diffusion models; data analysis and visualization.

Memristor/Devices: Memristor-array simulation; cleanroom fabrication; ALD, LPCVD, sputtering, ICP etching, RTP; C–V measurement; SEM analysis.

Hardware/Systems: Verilog; FPGA prototyping with Xilinx Nexys A7; VLA fine-tuning and robotic deployment.

Honors: Second Prize, South China Division, National College Student Integrated Circuit Innovation and Entrepreneurship Competition (Jul. 2025); Third Prize, FPGA Track, National Undergraduate Embedded Chip and System Design Contest (Dec. 2025).